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MILITARY STANDARD

THERMAL INSULATION REQUIREMENTS FOR MACHINERY AND PIPING

FSC 5640

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362

Thermal Insulation Requirements for Machinery and Piping

MIL-STD-769G(SH)

1. This Military Standard is approved for use by the Naval Sea Systems Command and is available for use by all Departments of Defense.

* 2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. This standard covers basic thermal insulation requirements. The information contained in this standard amplifies the generalized requirements for insulation of piping, machinery, uptakes, and mechanical equipment covered in the General Specifications for Ships of the U.S. Navy or in Ship Specifications for individual ships or classes of ships.

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1. SCOPE

1.1 The purpose of this standard is to prescribe the requirements for thermal insulation of piping, machinery, uptakes, and mechanical equipment for ships of the U.S. Navy.

2. REFERENCED DOCUMENTS

* 2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-775 - Steel Sheets, Carbon, Zinc-Coated.
QQ-W-343 - Wire, Electrical (Uninsulated).
SS-C-160 - Cements, Insulation, Thermal.
SS-C-1960 - Cement and Pozzolan, General Requirements for.
SS-C-1960/3 - Cement, Portland.
TT-P-320 - Pigment, Aluminum; Powder and Paste, for Paint.
UU-B-790 - Building Paper, Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant).
PPP-T-42 - Tape, Packaging/Masking, Paper.

MILITARY

MIL-T-2118 - Trap, Steam, Thermostatic, Naval Shipboard Use.
MIL-I-2781 - Insulation, Pipe, Thermal.
MIL-I-2818 - Insulation Blanket, Thermal, Fibrous Mineral.
MIL-I-2819 - Insulation Block, Thermal.
MIL-C-2861 - Cement, Insulation, High-Temperature.
MIL-A-3316 - Adhesives, Fire-Resistant, Thermal Insulation.
MIL-P-15280 - Plastic Material, Unicellular (Sheets and Tubes).
DOD-P-15328 - Primer, (Wash), Pretreatment, (Formula 117 for Metals).
MIL-I-15475 - Insulation Felt, Thermal, Fibrous Glass, Semirigid.
MIL-I-16411 - Insulation Felt, Thermal, Glass Fiber.
MIL-B-19564 - Bedding Compound, Thermal Insulation Pipe Covering.
MIL-C-19565 - Coating Compounds, Thermal Insulation Pipe Covering - Fire-, and Water-Resistant, Vapor-Barrier and Weather-Resistant.
MIL-C-20079 - Cloth, Glass; Tape, Textile, Glass; and Thread, Glass.
MIL-I-22023 - Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.
MIL-I-22344 - Insulation, Pipe, Thermal, Fibrous Glass.
MIL-C-22395 - Compound, End Sealing, Thermal Insulation Pipe Covering - Fire-, Water-, and Weather-Resistant.
MIL-I-23128 - Insulation Blanket, Thermal, Refractory Fiber, Flexible.
MIL-I-24244 - Insulation Materials, Thermal, With Special Corrosion and Chloride Requirements.
DOD-P-24555 - Paint, Aluminum, Heat-Resisting (650°C), Low-Emissivity (0.40 or Less).
DOD-E-24607 - Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin, Semigloss.

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DRAWINGS

NAVAL SEA SYSTEMS COMMAND
NAVSHIPS 803-2145518 - Spray Shield for Mechanical Joints.
NAVSHIPS 804-841336 - Piping Boiler Soot Blower.

(Copies of specifications and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip, Specification for. (DoD adopted)
B 209 - Aluminum-Alloy Sheet and Plate, Specification for. (Metric) (DoD adopted)
C 552 - Cellular Glass Block and Pipe Thermal Insulation, Specification for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 Application of the requirements listed herein is limited only to the equipment specified in 1.1.

3.2 The installation of asbestos or asbestos-containing insulation and lagging materials is not permitted. Where previously installed asbestos containing materials are removed, restoration shall be with asbestos-free materials as specified herein.

4. GENERAL REQUIREMENTS

4.1 Nominal thicknesses. Tables I to X, inclusive, specify only approved materials for insulation and lagging and the nominal acceptable thicknesses for the temperature ranges listed. The thicknesses specified in the following tables for hot surface insulation are designed to maintain the surface temperature at or below 125 degrees Fahrenheit (°F) for fluid temperatures up to 650°F with an ambient temperature of 85°F. For fluid temperatures above 650°F, the surface will be maintained at a maximum of 133°F.

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TABLE I. Insulation and lagging materials.[1/]

Service	Temperature range (°F)	Pipe and tubing		Valve and fittings		Flange joints		Machinery	
		Insulation	Lagging	Insulation	Lagging	Insulation	Lagging[2/]	Insulation	Lagging
Gases, steam, hot water, oil	125 to 1200	MIL-I-2781 MIL-C-20079[3/] MIL-I-16411 or MIL-I-23128[3/], class 2 MIL-I-22344 (370°F max) MIL-R-15280 (180°F max)	MIL-C-20079	MIL-I-2781 MIL-I-2819, class 2 MIL-I-16411 MIL-C-2861 SS-C-160 MIL-I-22344 (370°F max) MIL-P-15280 (180°F max) MIL-I-23128, class 2	MIL-C-20079	MIL-I-2781 MIL-I-2819, class 2 MIL-I-16411 MIL-C-2861 SS-C-160 MIL-I-22344 (370°F max) MIL-P-15280 (180°F max) MIL-I-23128, class 2	MIL-C-20079	MIL-I-2819, class 2 MIL-I-16411 MIL-I-2818 MIL-C-2861 SS-C-160 MIL-I-22023 (370°F max) MIL-I-23128, class 2	MIL-C-20079
Cold water Chilled water	28 to 70	MIL-I-16411 MIL-I-2781 MIL-I-22344 MIL-P-15280 ASTM C 552	MIL-C-20079 UU-B-790, type III, grade F, style 10	MIL-I-16411 MIL-I-2781 MIL-I-22344 MIL-I-2819, class 2 MIL-P-15280 ASTM C 552	MIL-C-20079 UU-B-790, type III, grade F, style 10	MIL-I-16411 MIL-I-2781 MIL-I-22344 MIL-I-2819, class 2 MIL-P-15280 ASTM C 552	MIL-C-20079 UU-B-790, type III, grade F, style 10	MIL-I-16411 MIL-I-22023 MIL-I-2819, class 2 MIL-P-15280 ASTM C 552	MIL-C-20079 UU-B-790, type III, grade F, style 10
Refrig-erant	minus 20 to 60	ASTM C 552 MIL-P-15280	MIL-C-20079	ASTM C 552 MIL-P-15280	MIL-C-20079	ASTM C 552 MIL-P-15280	MIL-C-20079	ASTM C 552 MIL-P-15280	MIL-C-20079

1/ Additional materials are covered in 4.5 (metal lagging); 6.1.4 (boiler uptakes); 6.2 (securing antisweat insulation); 6.4.1 (weather deck hot piping).
2/ Flammable liquid flanges shall not be lagged (see Drawing 803-2145518).
3/ Used only as a laminate construction consisting of a glass fabric outer jacket with a fibrous glass felt or refractory fiber felt insert. (An inner jacket of 0.008-inch crimped stainless steel mesh is used where pads or thermal insulation tape is needed and where the temperature of the hot surface is 450°F or above) (see table IV).

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TABLE II. Thickness of insulation for hot piping conforming to MIL-I-2781.

Nominal pipe size (inches)	Temperature range (°F)	Nominal thickness (inches)1/
		Total
1/2, 1-1/2	125 - 388	1-1/2
	389 - 750	2-1/2
	751 - 950	3
	951 - 1050	4
2, 2-1/2	125 - 338	1-1/2
	339 - 388	2
	389 - 900	3
	901 - 1050	4
3 through 4-1/2	125 - 338	1-1/2
	339 - 388	2-1/2
	389 - 500	3
	501 - 900	3-1/2
	901 - 950	4
	951 - 1050	4-1/22/
5, 6	125 - 338	1-1/2
	339 - 388	2-1/2
	389 - 750	3-1/2
	751 - 900	4
	901 - 950	4-1/22/
	951 - 1050	5-1/22/
7	125 - 338	1-1/2
	339 - 388	2-1/2
	389 - 750	3-1/2
	751 - 900	4
	901 - 950	4-1/22/
	951 - 1050	5-1/22/
8 and larger	125 - 338	1-1/2
	339 - 388	2-1/2
	389 - 500	3-1/2
	501 - 750	4
	751 - 900	4-1/22/
	901 - 950	52/
	951 - 1050	62/

1/ Does not include finishing cement.
2/ May be installed in multiple layers.

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* TABLE III. Thickness of insulation conforming to MIL-P-15280[1/] and MIL-I-22344, for hot piping.

Temperature range (°F)	Specification	Nominal thickness
		Inch
125 to 180	MIL-P-15280[1/] or MIL-I-22344	1/2
181 to 250	MIL-I-22344	1/2
251 to 300	MIL-I-22344	3/4
301 to 370	MIL-I-22344	1

1/ Approved for submarines only in this temperature range.

* TABLE IV. Thickness of insulating tape conforming to MIL-C-20079[1/], MIL-I-16411 or MIL-I-23128, class 2 for 1/4 to 3/4 inch size hot piping.

Temperature range (°F)	Pipe size	Nominal thickness
		Inch
125 to 250	1/4, 3/8	3/8
251 to 750	1/4, 3/8	1-1/2
125 to 250	1/2, 3/4	1/2
251 to 388	1/2, 3/4	1
389 to 500	1/2, 3/4	1-1/2
501 to 750	1/2, 3/4	2

1/ Used only as a laminate construction consisting of a glass fabric outer jacket with a fibrous glass felt or refractory fiber felt insert. (An inner jacket of 0.008-inch crimped stainless steel mesh is used where pads of thermal insulation tape are needed and where the temperature of the hot surface is 450°F or above.)

TABLE V. Thickness[1/] of insulating materials for hot surfaces of machinery and equipment up to 850°F.

	Nominal thickness (inches)		
Temperature range (°F)	Fibrous glass felt, MIL-I-16411, type II; or refractory fiber blanket, MIL-I-23128, grade A, class 2	Insulation, block, MIL-I-2819; or mineral fiber blanket, MIL-I-2818	Insulating cement MIL-C-2861
125 to 338	1	1-1/2	2
339 to 388	1-1/2	2	2-1/2
389 to 500	2	2-1/2	3-1/2
501 to 750	3	4	5
751 to 850	4	5	5-1/2

1/ Does not include finishing cement.

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TABLE VI. Thickness[1/] of insulating materials for hot surfaces of machinery and equipment over 850°F.

Temperature range (°F)	Nominal thickness (inches)	
	Single felt material	Block
	MIL-I-16411, type II, or MIL-I-23128, grade A, class 2	MIL-I-2819
851 to 950	4-1/2	5
951 to 1050	5	5-1/2
1051 to 1200	6	6-1/2

1/ Does not include finishing cement.

* TABLE VII. Thickness of refrigerant insulation for piping.

Pipe size (inches)	Temperature range (°F)	Cellular glass ASTM C 552 Nominal thickness (inches)		Plastic foam, MIL-P-15280 Nominal thickness (inches)	
Up to 1-1/4	Minus 20 to minus 1	2-1/2	1-1/2[1/]	1-1/2	1[1/]
	0 to 40	2	1-1/4[1/]	1	3/4[1/]
1-1/2 to 2-1/2	Minus 20 to minus 1	2-1/2	1-3/4[1/]	1-1/2	1[1/]
	0 to 40	2-1/2	1-1/2[1/]	1	3/4[1/]
3 to 5	Minus 20 to minus 1	3	2[1/]	1-1/2	1[1/]
	0 to 40	3	1-3/4[1/]	1	3/4[1/]

1/ Thickness for application in air-conditioned spaces only.

* TABLE VIII. Thickness of refrigerant insulation for machinery and equipment (exclusive of vapor barrier).

Temperature range (°F)	Nominal thickness (inches)			
	Foam plastic, MIL-P-15280		Cellular glass, ASTM C 552	
0 to 35	3	1[1/]	5	1-1/2[1/]

1/ Thickness for application in air-conditioned spaces only.

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* TABLE IX. Thickness of antisweat insulation (exclusive of vapor barrier).

Temperature range(°F)	Machinery and equipment		Piping		
	Material specification	Nominal thickness (inches)	Pipe size (inches)	Material specification	Nominal thickness (inches)
28 to 99	MIL-I-2819	1-1/2 3/4 1/	1/2 and above	MIL-I-2781	1 1/2 1/
	ASTM C 552	1 1/2 1/	1/4 and above	ASTM C 552	1 1/2 1/
	MIL-I-22023	1 1/2 1/	1/4 and above	MIL-P-15280	3/4 1/2 1/
	MIL-P-15280	3/4 1/2 1/	1/2 and above	MIL-I-22344	3/4 1/2 1/

1/ Thickness for application in air-conditioned spaces only.

* TABLE X. Nominal thicknesses of insulation for weather deck hot piping.

Pipe size (inches)	Calcium silicate, MIL-I-2781 Cellular glass, ASTM C 552
	Inches
1/4 to 3	1-1/2
3-1/2 to 6	2
Over 6	2-1/2

* 4.2 Special conditions. The following special conditions supplement or modify the selection of materials or thicknesses specified, when applicable:

(a) The insulation thickness on soot blower piping between the root valve and the soot blower heads shall be reduced to one half that indicated for a system normally at the same temperature.

(b) Where double layer construction is authorized as an alternate construction, insulation may be furnished in a single thickness equal to the total thickness specified, if single layer construction is considered desirable (see note 2, table II). Where single layer construction is used in lieu of double layer construction, suitable expansion joints to permit thermal movement of the piping, without opening of insulation joints, must be provided.

(c) Insulation conforming to MIL-I-2781 or cellular glass insulation conforming to ASTM C 552 shall be used on hot piping requiring insulation that will be exposed to the weather, and shall conform to the thickness specified in table X.

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(d) Elastomeric foamed plastic insulation, MIL-P-15280, may be used for machinery and equipment applications up to 180°F; 1/2 inch nominal thickness.

(e) Where HOT SURFACE insulation thicknesses are not specified, such as for refractory fiber insulation felt, MIL-I-23128, and special applications, the following shall be used as a guide in determining acceptable thicknesses:

(1) Insulation thickness shall be sufficient to reduce the insulation surface temperature to 125°F (see 4.1).

(f) Insulation materials (including insulation, adhesive and cement) used on austenitic stainless steel components and piping except for antisweat and refrigeration types shall meet the requirements of MIL-I-24244.

(g) Adhesives containing halogenated solvents shall not be used for submarine applications.

(h) Two feet of pipe immediately upstream of thermostatic steam traps MIL-T-2118 shall be insulated with 1/4 inch of insulation cement, SS-C-160, type III, grade F, and covered with lagging cloth. A removable cover made of two thicknesses of cloth shall be installed over the trap.

(i) Small diameter piping, 1 inch outside diameter (o.d.) and under shall not be insulated when the operating temperatures are between 125°F and 150°F.

(j) Shielding only shall be provided where such pipes are readily accessible to contact with personnel.

* 4.3 Adhesives. Adhesives conforming to MIL-A-3316 shall be used for fastening fibrous glass cloth and tape lagging. Lagging pretreated with compatible adhesive is acceptable providing the end result is equivalent.

4.4 Finishing cements. Where finishing cement is specified, any of the following materials are acceptable subject to any material limitations for the proposed application:

(a) Finishing cement, SS-C-160, type III, grade F.

(b) High-temperature insulating cement, MIL-C-2861, when used under fibrous glass cloth.

(c) A mixture of 80 percent high-temperature insulating cement, MIL-C-2861, and 20 percent Portland cement, SS-C-1960 and SS-C-1960/3.

4.5 Metal lagging. Where metal lagging is required, any of the following materials are acceptable, except for uptake applications (see 6.1.4):

Sheet material	Specification	Nominal thickness
Hot-dipped galvanized steel	QQ-S-775	0.014
Aluminum	ASTM B 209, 6061	.030
Corrosion-resistant steel (CRES)	ASTM A 167, type 304	.014

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4.6 Fasteners. Insulation shall be held in place by suitable wire or flat metal bands. The welding of fasteners to machinery, piping, pressure vessels, or other related equipment is prohibited. Where fasteners are necessary they shall be attached during manufacture (prior to heat treatment, stress relief, and testing) by a NAVSEA approved procedure.

5. DETAILED REQUIREMENTS

5.1 Hot-surface insulation covers. In order to insure that the pipe covering will not interfere with the servicing of a takedown joint where a reusable cover is installed, the permanent insulation shall stop short of the takedown joint and a short removable and reusable section of insulation shall be installed between the permanent insulation and the takedown joint. The insulation joint formed by the permanent and reusable sections may be square, or at an angle of 45 degrees; the joint, however, shall be tight, without any gaps between the two sections and shall incorporate means to prevent dislodging the insulation sections. Reusable covers are not required on systems insulated with elastomeric foamed plastic insulation conforming to MIL-P-15280.

5.2 Construction. For sizes larger than 2-inch iron pipe size (ips), valve bonnets and valves having takedown joints at the ends shall be fitted with reusable covers such that the bonnet joint may be removed independently of the valve covering. Valves, 2-inch ips and under, shall be fitted with separate covers as indicated above, or covers of a one-piece design such that they may be wrapped around the entire valve body and clipped or otherwise secured just below the handwheel.

5.3 Fabrication, piping components. For piping components except as otherwise specified, any one of the following methods of fabrication is acceptable:

* 5.3.1 Covers may be made in two half-section, using thermal insulating felt enclosed with 0.008-inch diameter knitted wire mesh on the inside and end surfaces; and with fibrous glass fabric conforming to MIL-C-20079, class 9, on the outside of the covers. Each half cover may be sewn and quilted with polytetrafluoroethylene (PTFE) coated fibrous glass yarn conforming to MIL-C-20079, type III, class 4, for hand sewing; or PTFE coated fibrous glass sewing thread (fully sintered) type III, class 3, for machine sewing. The covers may also be fastened with mechanical stapling with stainless steel staples in a manner to provide uniform thickness, strength and rigidity. Covers that are exposed to temperatures under 450°F may be constructed with fibrous glass fabric conforming to MIL-C-20079, class 9, for both inner and outer surfaces with a fibrous glass insert conforming to MIL-I-16411 or refractory fiber felt conforming to MIL-I-23128, class 2.

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* 5.3.1.1 For all covers exposed to temperatures 450°F and over, a 0.008-inch diameter knitted wire mesh shall be used on the inside surface and on the ends. Fibrous glass cloth conforming to MIL-C-20079, class 9, shall be used on all outside surfaces. Covers for use at temperatures up to 1200°F shall have a filling consisting of either fibrous glass felt, MIL-I-16411 or refractory fiber felt, MIL-I-23128, class 2. (See tables I, V and VI.)

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5.3.1.1.1 Knitted wire mesh shall be of 304 annealed stainless steel. The wire shall be 0.008-inch diameter. The mesh shall consist of 7-1/2 ± 1/2 courses per inch equal spacing and 10 ± 1 wales per inch equal spacing. The mesh shall be furnished in 30 ± 1/2 inch flattened tubular form and shall be crimped 0.125 to 0.150 inch deep by 5/16 inch crimp to crimp.

5.3.1.2 Preformed fibrous glass valve or fitting covers may be used when temperatures are in the 125°F to 370°F range. These shall be of the same thickness as the adjacent pipe covering. Such covers, when used, shall be lagged independently of the pipe covering and in a manner which will facilitate removal and replacement.

5.3.2 Covers may be made of segments of block insulation or preformed pipe insulation, having the same thickness as that on the adjacent piping. Blocks shall be securely wired to frames of 1/2-inch square mesh, number 18 gage (0.049-inch diameter) galvanized steel wire. Wire mesh frames inside and outside of blocks shall have ends bent over and joints secured with number 18 gage black annealed iron wire woven through the mesh. Insulating cement compatible with the material of the blocks shall be troweled smoothly over all surfaces of the wire mesh. Mineral wool roll felt, in accordance with MIL-I-2818, may be used to build up covers when the flange diameter is larger than the outside diameter of the adjacent pipe covering. Cover shall be tightly and smoothly lagged to envelop the outside and ends. Lagging shall be fibrous glass cloth conforming to MIL-C-20079 as described in 5.3.1.1. Lagging may be cemented or sewn on, except ends of covers which shall always be sewn. Where double layer insulation is used, the two sections of the cover shall be fitted together with scarfed joint. Such joints shall be straight and true to reduce heat loss. Bands, eyelets, or locks of galvanized steel, or lacing with hooks, rings, washers, and wire shall be used to secure the covers.

5.3.3 When installing the above covers, spaces between inner surfaces of covers for flanges and other irregular surfaces shall be filled with pieces of insulation felt. Felt shall be packed loosely enough to preserve air cell structure and tightly enough to prevent air circulation.

5.4 Fabrication, machinery and equipment. For reusable covers for machinery and equipment, either of the following methods of fabrication is acceptable.

5.4.1 Covers may be similar to the flexible refractory felt or fibrous glass felt type described for piping components.

5.4.2 Covers may be made in sections formed of insulating block held together with wire and adhesive cement, covered with 1/2-inch thickness of finishing cement, and lagged. Lacing with hooks, rings, washers, and wire, or brass snap fasteners shall be used to secure the covers.

* 5.4.3 Semi-removable turbine casing flange covers may be installed as an alternate for removable covers specified above. The permanent insulation shall be run to the casing flange allowing bolt removal space. The flange and bolts are then covered with mineral wool, wire inserted fibrous glass cloth, or wire

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mesh, as required by operating temperature, which shall be secured to the bolts with wire. The flange may then be insulated with fibrous glass felt, MIL-I-16411, mineral wool felt, MIL-I-2818, flexible refractory fiber felt, MIL-I-23128, class 2 or insulation block, MIL-I-2819 to the required thickness and shape; the insulation is then lagged with fibrous glass cloth. This cloth shall be carried over the outer edge of the permanent insulation and secured with adhesive. The semi-removable cover shall then be sealed and painted.

6. INSTALLATION

6.1 Hot surface insulation.

6.1.1 Pipe and tubing. Each layer of molded insulation shall be installed with joints butted together. Where two layers are used, joints shall be staggered. Not less than three fastenings shall be used for securing each 3-foot section of insulation. Fastening shall be number 18 gage minimum (0.049-inch diameter) annealed black or hot-dipped galvanized iron wire or flat steel bands. Except as otherwise specified, lagging shall be installed over the insulation.

* 6.1.1.1 The installation of soot blower piping insulation shall be in accordance with Drawing 804-841336.

6.1.2 Piping components. For valves, fittings, and accessories, welded and brazed fittings, including unions, may be insulated and lagged similarly to adjacent piping.

* 6.1.2.1 Block, felt, blanket insulating materials, or molded pipe insulation secured with hot-dipped galvanized iron or steel wire, may be used. When insulating felts are used, the inner layer shall be either fibrous glass felt conforming to MIL-I-16411 or refractory fiber felt, MIL-I-23128, class 2. Galvanized iron or steel wire netting, number 18 gage minimum (0.049-inch diameter), shall be spread over the insulating material and secured with wire. Insulating cement shall be used to fill crevices, smooth surfaces, and completely cover the wire netting. A 1/2-inch thickness of finishing cement shall then be applied. Alternatively, wire netting may be omitted where the size of the installation does not require netting to hold the insulation cement in place during the installation process. For these installations, glass cloth may be installed over the previously finished insulation material without the intermediate layer of wire mesh. Insulating material shall be the same thickness as that on adjacent piping.

6.1.2.2 For components 3-1/2 inch ips and smaller, insulating cement only conforming to MIL-C-2861 may be applied to a thickness 1/2 inch less than the adjacent pipe insulation. A 1/2-inch thickness of finishing cement shall be applied over the insulating cement.

6.1.2.3. Reusable covers shall be fitted where required.

6.1.3 Machinery and equipment. For machinery and equipment, block, felt, or blanket insulating materials of the required thickness shall be secured with hot-dipped galvanized iron wire. Galvanized iron wire netting 1 inch mesh and number 18 gage minimum (0.049-inch diameter) shall be spread over the surface and secured by wire. Insulating cement shall be used to fill crevices, smooth surfaces, and completely cover the wire netting.

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6.1.3.1 When no insulating cement has been specified, a 1/2-inch thickness of finishing cement shall be applied.

6.1.3.2 When an insulating cement has been specified it shall be applied in successive layers, 1/2-inch to 1 inch in thickness, until the total thickness specified has been reached. Wire netting, similar to that used for covering the insulating materials shall be installed between layers. A 1/2-inch thickness of finishing cement shall be applied over the last layer of insulating cement.

6.1.3.3 Lagging shall be installed over finishing cement. Reusable covers shall be installed where required.

6.1.3.4 Clips, hooks, or other fastenings for securing insulation or lagging shall not be brazed or welded to nonferrous parts of distilling plants or deaerating feed tanks.

6.1.4 Boiler uptakes. Boiler uptake thermal insulation shall be 2 inches thick. Either mineral wool felt, MIL-I-2818, or fibrous glass sheet, MIL-I-15475, may be used. If acoustic absorptive treatment is found to be necessary to decrease the noise level, the insulation thickness shall be increased accordingly.

6.1.4.1 Metal lagging for uptakes shall be galvanized sheet steel conforming to QQ-S-775, not less than 1/32 inch thick.

6.1.4.2 Insulation and lagging is not required on uptakes above the weather deck, except where the transfer of heat to spaces adjacent to the uptake area would be objectionable.

6.1.5 Unfired pressure vessels. Unfired pressure vessels, including catapult wet accumulators, shall be covered with block insulation, MIL-I-2819, in accordance with table V. Block insulation shall be covered with 1/2-inch cement, MIL-C-2861, lagged with fibrous glass cloth, MIL-C-20079, type I, class (as applicable), and painted in accordance with 6.6. Insulation in the way of vessel supports shall be metal faced to prevent insulation from wedging between the vessel and its support.

6.1.5.1 Removable and reusable covers shall be installed over butt welded shell inserts for which periodic radiographic inspection of the joint is required. These covers shall extend 4 inches beyond the welded joint.

6.1.6 Outer boiler casing. Insulation, conforming to MIL-I-2819, class 2, shall be secured to casing by wire netting (number 18 gage) laced to welded notched studs on boiler casing. Insulating cement, SS-C-160, shall be used to fill crevices, smooth surfaces and completely cover the netting to 1/2 inch thickness. Glass cloth conforming to MIL-C-20079, type I, class (as applicable), shall be used to lag the insulation and shall be painted in accordance with 6.6.

6.2 Antisweat insulation (cold and chilled water service).

6.2.1 Preformed pipe covering shall be secured to the pipe in the manner prescribed in 6.1.1. Fibrous glass felt insulation shall be secured with number 18 gage minimum (0.049-inch diameter) hot-dipped galvanized iron wire, soft annealed copper wire, QQ-W-343, wire inserted fibrous glass yarn, or glass thread, MIL-C-20079, type III, class (as applicable), spirally wound on 1-inch centers.

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One layer of water repellent and fire resistant paper, UU-B-790, shall be wrapped tightly around the insulation and secured with glass thread, MIL-C-20079, type III, class (as applicable), or 1-inch wide tape, PPP-T-42. All joints of the paper shall be lapped and sealed with adhesive cement, MIL-A-3316, class 1. The compatible lagging shall then be installed and completely covered with vapor barrier compound, MIL-C-19565, type II. The water repellent paper may be omitted on cellular glass where the insulation surface is suitable for the effective application of vapor barrier compound MIL-C-19565.

6.2.2 Application of a vapor barrier is not required on elastomeric foamed plastic insulation, MIL-P-15280. Lagging shall be applied to protect insulation from damage, in high traffic areas.

6.3 Refrigerant insulation.

* 6.3.1 Cellular glass insulation shall be coated on all surfaces with vapor barrier compound, MIL-C-19565, type II, at the time of installation. Insulation shall be installed with staggered end joints. On horizontal pipes the longitudinal joints shall be at the top and bottom. Insulation shall be secured with 1/2 inch wide metal bonds or 1-inch wide tape, PPP-T-42 on 9-inch centers. The compatible lagging shall then be installed and completely covered with vapor barrier compound, MIL-C-19565, type II.

6.3.2 Elastomeric foamed plastic, MIL-P-15280, may be applied in 1/4-inch minimum thickness layers as necessary to build up the required thickness (form T or S, as applicable). All longitudinal and butt joints shall be staggered. All joints and lagging, if required (see 6.2.2), shall be secured in accordance with adhesive cement requirements of MIL-P-15280.

6.4 Weather deck hot piping insulation.

* 6.4.1 Calcium silicate or cellular glass insulation for piping exposed to the weather shall be installed as follows:

(a) Preliminary preparation of piping.
  (1) All surfaces to be clean, dry, and free of scale and grease.
  (2) Fittings, valves, flanges, pipe supporting clamp, and at least 3 inches of adjacent pipe shall be painted as follows: Apply one coat pretreatment formula 117, DOD-C-15328 to a maximum dry film thickness (DFT) of 0.0005 inch (0.5 mil). After this coat dries, apply two coats of aluminum paint made by mixing 2 pounds of aluminum paste, TT-P-320, type II, class B, with each gallon of phenolic varnish for temperatures up to 300°F. Above 300°F, apply two coats of paint conforming to DOD-P-24555.

(b) Installation on pipes.
  (1) The bore, butt ends, and longitudinal joint surfaces of cellular glass insulating material shall be coated not more than 1/16 inch thick with commercial bedding compound, in accordance with MIL-I-19564, at time of installation. Bedding compound is not required with calcium silicate pipe covering.
  (2) Longitudinal joints on horizontal piping shall be on top and bottom of pipe.

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(3) Insulation shall be secured tightly to pipe with 1/2 inch wide 22 gage galvanized steel bands on 9-inch centers. Steel bands shall be placed over a layer of fibrous glass tape, MIL-C-20079, type II, class (as applicable), which has been dipped in the commercial finishing compound in accordance with MIL-C-19565, type I. Steel bands shall be wrapped with a layer of masking tape, PPP-T-42.

(4) Completely coat insulation with commercial finishing compound, in accordance with MIL-C-19565, using about 2 gallons per 100 square feet. Wrap tightly with one layer of knitted fibrous glass tape, MIL-C-20079, type II, class 3, and then apply another coating of above-specified finishing compound, using about 4 gallons per 100 square feet. After this coat has set, apply a second coat of finishing compound using the same quantities.

(5) Where insulation is stopped off on the piping, sufficient mineral wool, MIL-I-2818, shall be tightly tied in place with galvanized iron wire over a heavy coating of the above-specified commercial bedding compound, to provide a tapered portion from insulation surface to pipe surface. The ends of the insulation shall be tapered at a 30-degree (approximately) angle with the pipe. The tapered ends of the insulation shall be smoothed with insulation cement in accordance with MIL-C-2861. The cement covered tapered ends, after drying thoroughly, shall be coated with approximately an 1/8-inch thick tack coat of end sealing compound in accordance with MIL-C-22395. The sealer compound shall extend onto the pipe for at least 3 inches. A single layer of fibrous glass cloth lagging, in accordance with MIL-C-20079, type I, class (as applicable), shall be applied over the insulation and secured at longitudinal lap joint with class 1 adhesive cement in accordance with MIL-C-3316. The fibrous glass lagging cloth shall be tailored to fit the contour of the ends of the insulation by cutting and removing wedge-shaped sections of the cloth. The remaining ends of the cloth shall be embedded in the tack coating of sealer compound and shall be attached to the pipe with a single 1/2-inch wide galvanized steel band. A 3/16-inch layer (approximately) of sealer compound shall be troweled to a smooth finish over the cloth covered ends of the assembly. A smooth finish may be obtained by brush coating or hand rubbing the sealer compound with a suitable solvent. After 72 hours of drying at ambient temperature, the fibrous glass cloth of the assembly shall be given two brush coats of water- and weather-resistant coating compound in accordance with MIL-C-19565. The water-proofing compound shall extend halfway down the tapered ends of the assembly. The water-proofing compound shall be air dried 24 hours between applications. As an alternate method, the preformed pipe insulation may be tapered, the exposed surfaces coated with insulation cement, SS-C-160, type III, grade F, and lagging and sealant applied as described above.

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(c) Installation on fittings, flanges, and valves.

(1) Before applying flange insulation, weather deck piping shall be tested and secured in the following manner: After specified tests are completed, weather deck piping shall be subjected to alternate periods of full operating pressure, allowing pipe to come to maximum temperature; and then to zero gage pressure allowing pipe to come to ambient temperature. These cycles shall be repeated a sufficient number of times tightening and adjusting flanges where necessary until no leaks can be detected.

(2) Fittings, flanges, and valve covers shall be ship-fabricated from sections of molded pipe covering or cellular glass block cemented together with Amchem Products, Inc., Foster Division, 82-48 (fast setting) adhesive or equal.

(3) Permanent covers for fittings and valves shall be fitted snugly to fittings and adjacent pipe covering using the same material and methods as outlined for pipe covering. Voids between insulation and fitting shall be filled with tightly packed mineral wool, MIL-I-2818.

(4) Where specified, rigid-type portable flange covers shall extend over the adjacent pipe covering 1-1/2 times the thickness of the insulation. The two halves of the cover should be coated and lagged separately, using the same materials and procedure as outlined for pipe covering. The galvanized steel bands used to secure the two halves together and to the adjacent pipe covering shall be applied over the lagging and then coated with the above-specified finishing compound.

(d) Installation around supports and hangers.

(1) Remove only enough insulation from butt edges to provide a snug fit around support brackets or hanger rods. Fill voids between insulation and support with tightly packed mineral wool, MIL-I-2818, to within 1/4 inch from insulation surface. Fill remainder of space with end sealing compound, in accordance with MIL-C-22395, overlapping generously both the support member and the adjacent insulation. Lag and coat with the same method and materials as adjacent piping.

6.5 Metal lagging. Metal lagging, where required, shall be installed with lap joints, secured with hardened self-tapping screws or metal bands. Joints shall be arranged in a manner which will facilitate run-off of impinging liquids.

* 6.6 Painting. Cloth and tape lagging shall be painted after installation with one coat of nonflaming paint conforming to formulas no. 124 of DOD-E-24607, if necessary for appearance. Elastomeric insulation conforming to MIL-P-15280 shall be painted as follows to provide improved fire resistance; apply one coat of Ocean Chemical Co., "Ocean 634" 1 to 2 mils DFT followed by two coats of "Ocean 9788" 5 mils DFT per coat. Approved NAVSEA equivalent coating systems may also be used. The final coat of "Ocean 9788" may be topcoated to match surrounding structures with one coat of the appropriate color conforming to DOD-E-24607.

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6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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